Customer Churn Analysis

1. **Overall Churn Rate:**
    - The dataset indicates a **churn rate of 47.37%**, meaning nearly half of the customers have discontinued their service.
    - Understanding the characteristics of these churned customers is crucial for formulating retention strategies.
2. **Demographics & Churn Trends:**
    - **Gender Distribution:** Both male and female customers show a comparable churn pattern, with no significant gender-based difference.
    - **Tenure Impact:** Customers with **shorter tenure (0–12 months) exhibit a higher churn rate**, while long-term customers tend to stay.
3. **Service Usage & Payment Behavior:**
    - **Support Calls:** Customers making **6 or more support calls** show an increased likelihood of churn, indicating possible dissatisfaction.
    - **Usage Frequency:** Customers with inconsistent usage patterns tend to churn more frequently.
    - **Payment Delays:** More than **30% of churned customers have experienced frequent payment delays**, suggesting that financial constraints or billing issues contribute to churn.
4. **Subscription & Contract Type:**
    - Customers on **monthly contracts have the highest churn rate (~60%)**, while those with annual or quarterly contracts are more likely to stay.
    - The **Basic subscription plan** sees higher churn, while Premium users tend to remain engaged.

Business Implications & Recommendations:

- **Retention Strategies:** Target customers with short tenure and frequent support calls by offering personalized assistance and incentives.
- **Billing Improvements:** Address payment delays by offering flexible payment plans and automated reminders.
- **Subscription & Contract Adjustments:** Encourage long-term contract sign-ups through discounts and added benefits.
- **Customer Engagement:** Proactively engage with customers showing irregular usage patterns to improve service utilization.

This analysis highlights critical churn drivers and provides actionable recommendations to enhance customer retention and business sustainability